

Netflix Dataset Analysis: Exploratory Data Analysis (EDA)

This project presents an Exploratory Data Analysis (EDA) of the Netflix Titles dataset to identify trends and patterns in content distribution across content types, release years, countries, and genres. The original dataset contained 8,807 records and 12 variables, including key features such as type, country, release_year, rating, and listed_in (genres). Initial data inspection revealed missing values, particularly in the director, cast, and country columns. To improve data quality and ensure consistency, records containing missing values were removed, resulting in a cleaned dataset of 5,332 entries that was subsequently prepared for analysis.

The analysis shows that movies make up the majority of content in the cleaned dataset, while TV shows account for a much smaller proportion. This indicates that Netflix's catalog is largely movie-focused, although the data-cleaning process may have contributed to this imbalance. Analysis of release trends reveals a sharp increase in content additions after 2015, with activity peaking around 2019–2020. This growth reflects Netflix's rapid global expansion and increased investment in both original and licensed content. Furthermore, the release-year distribution is heavily concentrated after 2000, demonstrating a strong focus on modern entertainment, while older titles appear only occasionally.

Country-level analysis highlights the United States as the dominant source of Netflix content, contributing the largest number of titles by a significant margin. India ranks second, while countries such as the United Kingdom, Canada, Spain, and Nigeria contribute considerably fewer titles. This distribution suggests that Netflix's content library is concentrated in a small number of major production markets, with the United States playing a leading role.

Genre analysis indicates that Drama and Comedy are the most common categories within the dataset. Popular genre combinations such as *Dramas*, *International Movies* and *Stand-Up Comedy* occur frequently, showing Netflix's strategy of offering diverse and broadly appealing content. The prominence of international and multi-genre titles also reflects the platform's effort to serve a global audience with varied viewing preferences.

In conclusion, the Netflix dataset reveals several important patterns: the dominance of movies over TV shows, significant growth in content during the late 2010s, concentration of production in a few countries—particularly the United States—and the popularity of drama- and comedy-based genres. These findings provide valuable insight into Netflix's content strategy and demonstrate how exploratory data analysis can be used to uncover meaningful trends within large entertainment datasets.